

DISSERTATION PROPOSAL



Efficient Simulation of Non-Markovian Dynamics of Open Quantum Systems on Quantum Computers with a Focus on Quantum Tomography

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1 Motivation

Classical computers store and process information in the form of bits, which can only exist in one of two states, $|0\rangle$ or $|1\rangle$. Quantum computers, by contrast, use qubits, which, due to the principle of superposition, can exist in a combination of the classical states $|0\rangle$ and $|1\rangle$. A register of n qubits spans a 2^n -dimensional Hilbert space, meaning that a quantum state is described by 2^n complex amplitudes. This exponential growth of the state space is the main reason why quantum computing is considered so powerful: quantum devices can naturally encode and manipulate information that would require exponentially large memory to store on a classical machine. For example, a system of 500 qubits already spans 2^{500} basis states of the form $|x_1x_2\dots x_{500}\rangle$ with 2^{500} coefficients, where $x_1, \dots, x_{500} \in \{0, 1\}$. This number exceeds the estimated number of particles in the observable universe, demonstrating the enormous potential of quantum computing [1].

In practice, quantum computers do not replace classical processors entirely, but instead work in close synergy with them. As schematically illustrated in Fig. 1.1, a classical processor is essential for translating algorithms into sequences of quantum gates, controlling their timing and execution on the quantum hardware, and collecting the measurement data.

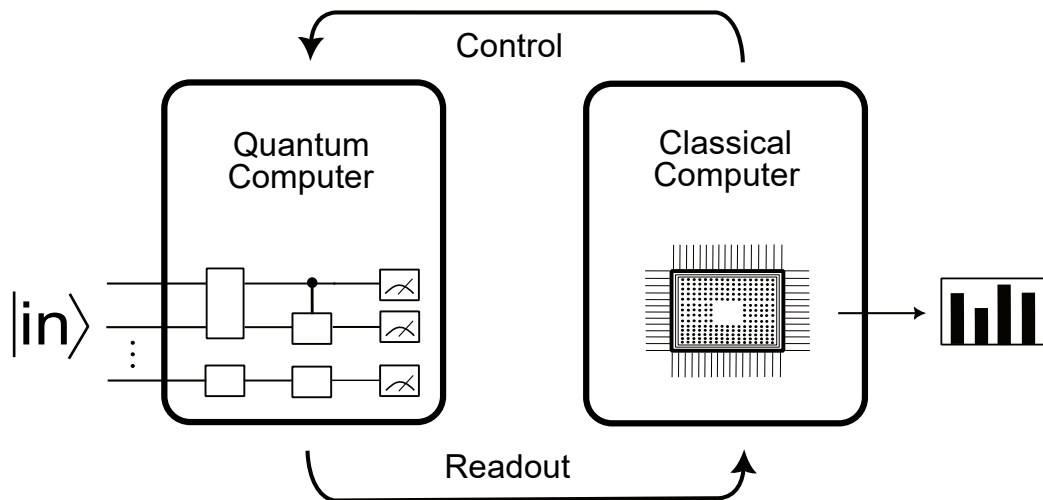


Figure 1.1: Synergy between classical and quantum computers. A quantum computer executes a programmable sequence of elementary quantum operations on an initial state, producing a final state that can then be measured. While the quantum computer performs the core calculations, a classical computer is required to decompose algorithms into sequences of quantum gates and collect the measurement data.

This division of labor allows the quantum processor to explore the exponentially large Hilbert space, while the classical processor guides the computation. Therefore, quantum computers are particularly promising for simulating open quantum systems, where the exponential growth of the state space quickly renders classical methods intractable [2].

2 Problem Definition and Current State of Research

Many systems of central scientific interest are inherently quantum mechanical, with state spaces that grow exponentially with system size. This exponential scaling makes exact brute-force simulations on classical supercomputers infeasible even for moderately sized systems [3]. Richard Feynman first articulated this observation in 1982, arguing that a controllable quantum device could simulate quantum dynamics more naturally and efficiently than any classical computer [4].

Recent research has developed algorithms to simulate the non-unitary dynamics of open quantum systems on quantum hardware. Since open systems interact with their environment, their dynamics are inherently non-unitary, whereas quantum computers natively support only unitary operations that preserve the norm of the qubits [5]. To bridge this gap, several methods use the concept of dilation, embedding a non-unitary matrix into a higher-dimensional unitary matrix that can be implemented in a quantum circuit. Foundational contributions in this direction include the works of Seneviratne et al. (2024) [6], Hu et al. (2022) [7], and Dan et al. (2025) [8], which demonstrate the theoretical feasibility of the methods.

Despite these advances, current methods remain largely on the proof-of-concept level. They rely heavily on classical computation, where the classical processor recomputes circuit descriptions, intermediate operators, or state updates at every time step. As a result, the entire time evolution must be simulated on a classical device first before quantum hardware can be involved, limiting the practical impact of existing algorithms.

Motivated by this shortcoming, I developed an enhanced algorithm during my Master's thesis that is designed to transfer a significant part of the computational workload from the classical processor to the quantum device and thus realizes the original promise that quantum computers can simulate quantum dynamics independently and more naturally. The key idea is to reuse the quantum state obtained at one time step as the input for the subsequent step, as illustrated in Figure 2.1. This approach enables the same circuit to be applied repeatedly for each propagation interval Δt , allowing the required circuit and corresponding operators to be computed only once during an initial preprocessing stage. After this setup, the quantum processor can carry out the entire time evolution without further classical intervention. The algorithm has been validated for Markovian dynamics, demonstrating that operations

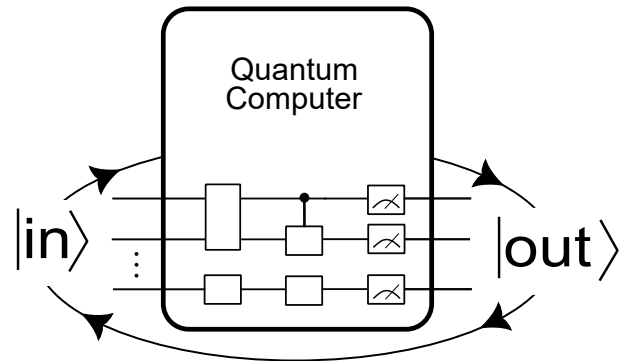


Figure 2.1: Efficient quantum state reuse across time steps without repeated intermediate classical processing at every step.

previously requiring intensive classical computation can instead be executed almost entirely on the quantum hardware. There is strong evidence that the same method can be extended to non-Markovian dynamics as well, although a formal proof of this generalization remains an open task.

The main caveat of the enhanced algorithm is that it depends on full quantum state reconstruction after each iteration in order to provide the correct input for the next circuit iteration.

In practice, after each time step Δt , the complete quantum state must be reconstructed. However, simple projective measurements yield only the probabilities of the basis states, that is, the squared magnitudes of the amplitudes, and do not provide the phase information. Because phase information is essential to correctly reusing the quantum state as input for the subsequent step, projective measurements in a single basis are insufficient. To address this shortcoming, quantum state tomography (QST) is employed, which measures in multiple bases to reconstruct amplitudes and phases and thus recovers the full quantum state.

However, quantum state tomography faces severe limitations that currently prevent straightforward scaling of the method. Most importantly, its resource requirements grow rapidly with the dimension of the Hilbert space. The number of measurements needed for a complete reconstruction scales as $\mathcal{O}(d^2)$, where d is the Hilbert space dimension. Although the scaling is polynomial in d , the dimension of an n -qubit system grows exponentially as $d = 2^n$, making the overall resource cost scale as $\mathcal{O}(4^n)$, i.e., exponentially in n [9]. This rapid growth makes full tomography impractical even for systems of moderate size. Promising strategies to overcome this limitation include matrix product state tomography [10] (2020), which reconstructs weakly entangled quantum states by learning local tensors that efficiently encode the global state; compressed sensing tomography [11] (2022), which exploits the low-rank structure of nearly pure states to reconstruct them from significantly fewer measurements; and machine-learning-based representations [12] (2022), which generalize from limited experimental data to obtain full quantum states.

In parallel, the limitations of current noisy intermediate-scale quantum (NISQ) devices must be taken into account. These systems are still prone to noise, have limited coherence times, and can only reliably execute circuits of modest depth. To ensure practical feasibility, the circuits implementing the effective dynamics must therefore be optimized for minimal depth and gate count. By combining efficient state-reconstruction methods with hardware-conscious circuit design, the proposed approach aims to enable time evolutions to be computed almost entirely on the quantum processor, ensuring that the main computational workload truly resides on the quantum hardware rather than on the classical processor.

3 Research Infrastructure and Expertise at JKU Linz

The Johannes Kepler University (JKU) Linz provides an excellent environment for my dissertation project, combining strengths in both theoretical and experimental quantum science. In the group of Univ.-Prof. Dr. Armando Rastelli, I have already gained hands-on experience with quantum dots and quantum optics, which will help me understand and control quantum systems and support my work in quantum information. At the same time, the Institute for Theoretical Physics, Department of Quantum Dynamics and Classical Dynamics, is led by Univ.-Prof. Dr. Tobias Kramer. His expertise in open quantum systems [13][14] and non-Markovian dynamics, particularly in the efficient implementation of the hierarchical equations of motion (HEOM) for complex simulations [15][16], provides an ideal environment for advancing my dissertation.

Several key resources at JKU make it particularly well-suited for my research:

- Substantial investment in the MUSICA supercomputer, coming online in fall 2025, enables efficient classical simulations and algorithm benchmarking.
- The membership in the Austrian Computing Alliance provides access to national computing infrastructure.
- Active FWF projects, such as the “Open Quantum Dynamics Lab”, encourage collaborations and knowledge exchange.
- The integration in *quantA*, Austria’s national quantum research cluster, connects to a wider network and the latest developments in the field.

The JKU is also home to the Institute for Integrated Circuits and Quantum Computing led by Univ.-Prof. Dr. Richard Küng, who is internationally recognized for his work in quantum information [17], quantum machine learning [18] and quantum state reconstruction [19][20]. This combination of expertise in quantum information, along with the modern facilities and collaborative environment at JKU, provides a strong foundation for my research. The university’s support for interdisciplinary projects in quantum computing ensures access to the tools needed to advance cutting-edge research.

4 Detailed Problem Formulation

Open quantum systems interact with, and are influenced by, their external environment. They appear across a broad spectrum of physical phenomena, from biological processes to applications in quantum technologies. Since these systems are not isolated, they cannot be described by a purely unitary evolution, as in the case of closed systems. Instead, their dynamics follow a non-unitary evolution. Here, a distinction is made between Markovian and non-Markovian dynamics. Markovian models neglect the memory effects of the environment and assume that the evolution of the system depends solely on its most recent state. This approximation is only valid for weak coupling between the system and the environment. In contrast, non-Markovian dynamics account for a more realistic environment retaining a memory of the system, leading to a more accurate but also more complex description of the dynamics [21].

This section explains the procedure for implementing the non-unitary dynamics of open quantum systems on quantum computers. The implementation of Markovian dynamics on quantum circuits is introduced first, following the approaches of Seneviratne et al. (2024) [6] and Hu et al. (2022) [7]. Subsequently, non-Markovian dynamics are discussed, as presented in Seneviratne et al. (2024) [6] and Dan et al. (2025) [8].

After introducing the methods, an enhanced algorithm developed in the context of my Master's work is presented, and its limitations and potential improvements are discussed.

4.1 Markovian Dynamics

The goal is to construct a quantum circuit to calculate the Markovian dynamics of open quantum systems on a quantum computer that only allows for unitary operations. To this end, first, the Markovian Lindblad equation is introduced:

$$\frac{d}{dt}\rho_S(t) = -\frac{i}{\hbar}[H, \rho_S(t)] + \sum_i \gamma_i (L_i \rho_S(t) L_i^\dagger - \frac{1}{2}\{L_i^\dagger L_i, \rho_S(t)\}) \equiv \mathfrak{L}\rho_S(t), \quad (4.1)$$

where $\mathfrak{L}$ is the so-called Lindbladian. The Lindblad equation represents the most general form of a Markovian master equation. It describes the time evolution of the system density matrix ρ_S as a result of interactions with the environment. Hence, it is crucial for describing non-unitary processes, like dissipation and decoherence. The unitary part of the Lindblad equation consists of the commutator of the Hamiltonian H with the system density matrix. The dissipative part involves jump or Lindblad operators L_i , which induce transitions, which are associated with decay rates γ_i . Each jump operator corresponds to a specific physical process that the system undergoes due to its interaction with the environment [22].

It can be shown that the Lindblad equation can be interpreted as a quantum map with so-called Kraus operators $M_k(t)$. This quantum map can be written as an operator sum representation:

$$\rho_S(t) = \sum_k M_k(t) \rho_S(0) M_k^\dagger(t). \quad (4.2)$$

The Kraus operator sum representation is a general formalism that describes the time evolution of quantum systems. It applies to both Markovian and non-Markovian dynamics, depending on the nature of the Kraus operators. The representation is not unique, and hence different sets of Kraus operators can describe the same dynamics. The Kraus operator sum representation can also be written as a quantum map or quantum channel Φ , with $\rho_S(t) = \Phi[\rho_S(0)]$, that propagates the density matrix of a system in time. In fact, Kraus' theorem states that any completely-positive trace-preserving map can be expressed as an operator sum representation [23].

4.1.1 Evaluation of the Kraus Operators

To implement the Lindblad equation on a quantum circuit, a vectorization Ansatz is required to arrive at the vectorized Lindblad equation:

$$\begin{aligned} \frac{d}{dt} \text{vec}(\rho_S(t)) = & \left[-\frac{i}{\hbar}(\mathbb{I} \otimes H) - \frac{i}{\hbar}(H^T \otimes \mathbb{I}) + \sum_i \gamma_i \left((L_i^* \otimes L_i) \right. \right. \\ & \left. \left. - \frac{1}{2}(\mathbb{I} \otimes L_i^\dagger L_i) - \frac{1}{2}(L_i^T L_i^* \otimes \mathbb{I}) \right) \right] \text{vec}(\rho_S(t)) = \mathbb{L} \text{vec}(\rho_S(t)). \end{aligned} \quad (4.3)$$

where $\mathbb{I}$ is the identity operation. This differential Equation can be solved with an exponential Ansatz, and yields an equation for the time-dependent vectorized density matrix:

$$\text{vec}(\rho_S(t)) = e^{\mathbb{L}t} \text{vec}(\rho_S(0)) = \mathcal{L}(t) \text{vec}(\rho_S(0)). \quad (4.4)$$

If $\mathbb{L}$ is time-independent, the time step for the time evolution can be arbitrarily large, as is expressed in the exponent. It can be shown that the superoperator $\mathcal{L}(t)$ is a completely-positive trace-preserving quantum map propagating $\rho_S(0)$ to $\rho_S(t)$. Hence, an operator sum representation can be found for $\mathcal{L}(t)$ where the corresponding Kraus operators are obtained using a so-called Choi decomposition. To this end, the Choi matrix is constructed using the involution operation acting on $\mathcal{L}(t)$:

$$C(t) := \sum_{i,j=1}^N (E_{i,j} \otimes \mathbb{I}) \mathcal{L}(t) (\mathbb{I} \otimes E_{i,j}). \quad (4.5)$$

Here, $\mathbb{I}$ is an $N \times N$ matrix, where N equals the number of rows or columns of H . Also, $E_{i,j}$ is an $N \times N$ matrix with a 1 in the $(ij)^{\text{th}}$ component, and 0 elsewhere.

Since $\mathcal{L}(t)$ is a completely-positive trace-preserving map, the Choi matrix is guaranteed to be positive and Hermitian, which ensures that it has positive eigenvalues. The Kraus operators corresponding to $\mathcal{L}(t)$ can be found by performing an eigen decomposition of the Choi matrix:

$$C(t) = \sum_{k=1}^{N^2} \sigma_k(t) u_k(t) u_k^\dagger(t), \quad (4.6)$$

where $\sigma_k(t)$ are the eigenvalues, and $u_k(t)$ are the eigenvectors of the Choi matrix at time t . These can be used to find the vectorized Kraus operators $M_k(t)$:

$$\text{vec}(M_k(t)) = \sqrt{\sigma_k(t)} u_k(t). \quad (4.7)$$

To obtain a Kraus operator of the correct dimensions, $\text{vec}(M_k(t))$ must be reshaped into matrix form again. The time evolution of the density matrix can then be calculated according to Equation 4.2, and the number of Kraus operators equals the number of eigenvectors of $C(t)$.

4.1.2 Singular Value Decomposition and Dilation

Quantum circuits are limited to unitary operations. Since the Kraus operators in Equation 4.2 represent the non-unitary evolution of an open quantum system, they are inherently non-unitary and must be dilated to unitary operators before being implemented on a quantum circuit.

To this end, first a singular value decomposition (SVD) is employed, which exploits that every real or complex matrix can be factorized into a rotation, a rescaling, and a second rotation. Performing the singular value decomposition on the Kraus operator $M_k(t)$ yields:

$$M_k(t) = U \Sigma V^\dagger. \quad (4.8)$$

Here, U , V , and Σ have the same dimensions as $M_k(t)$ and, in principle, depend on time, since they differ for each $M_k(t)$. However, the time argument will be omitted for brevity. The operators U and V correspond to the rotations, and hence are already unitary. The non-unitary part of $M_k(t)$ is encoded into Σ , which corresponds to a rescaling, and is therefore a non-unitary diagonal matrix. Thus, to implement Σ , it must be dilated into a unitary matrix:

$$U_\Sigma = \begin{pmatrix} \Sigma_+ & 0 \\ 0 & \Sigma_- \end{pmatrix}, \quad (4.9)$$

where $\Sigma_{\pm jj} = \sigma_{jj} \pm i\sqrt{1 - \sigma_{jj}^2}$, with σ_{jj} being the diagonal elements of Σ . The Σ_{+jj} and Σ_{-jj} are constructed for each diagonal element σ_{jj} of Σ , and hence the resulting unitary U_Σ is twice as large as Σ , $M_k(t)$, U , and $V^\dagger$. Since $\mathcal{L}(t)$ is a completely-positive trace-preserving map, $|\sigma_{jj}|$ is always smaller than 1, preventing negative expressions in the square root.

4.1.3 Construction of the Circuit

The operators $V^\dagger$, U_Σ , and U , obtained from the singular value decomposition, can be used to construct a quantum circuit, as shown in Figure 4.1. The aim of the circuit is to apply the Kraus operator $M_k(t)$ to the input state $|\psi\rangle$, which gives $M_k(t)|\psi\rangle$ as a result. Taking the outer product with itself yields $M_k(t)|\psi\rangle\langle\psi|M_k(t) = M_k(t)\rho_S M_k(t)$, which corresponds to one summand in the operator sum representation of Equation 4.2. Thus, it is expected that the circuit shares similarities with the singular value decomposition $M_k(t) = U\Sigma V^\dagger$, as it aims to calculate $M_k(t)|\psi\rangle$ in a unitary manner.

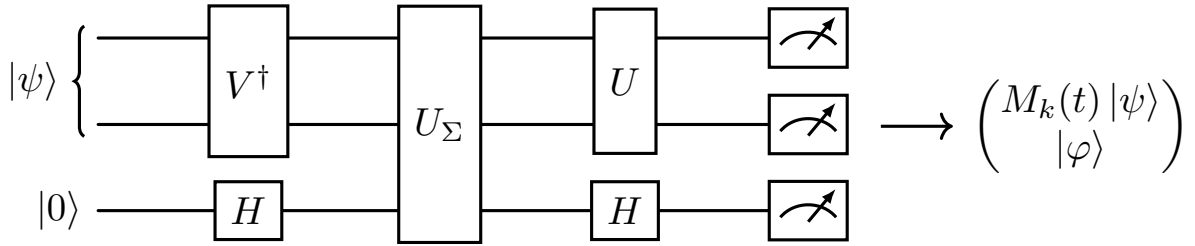


Figure 4.1: Quantum circuit for the calculation of the time dependence of the system density matrix. The operators $V^\dagger$, U , and U_Σ stem from the singular value decomposition. Applying them to the circuits enables a non-unitary evolution on a unitary quantum circuit. The result of the circuit, given an input state $|\psi\rangle$, is an output state vector whose first half yields the desired outcome $M_k(t)|\psi\rangle$. Repeating the calculation for all Kraus operators $M_k(t)$ and summing the results yields the correct time evolution of the system density matrix.

In general, the circuit consists of $n + 1$ qubits, where the Kraus operator, and thus also the system Hamiltonian and system density matrix, are of dimension $2^n \times 2^n$. The first n qubits implement the initial state $|\psi\rangle$ of the system, and thus correspond to the size of the system Hilbert space. The $(n + 1)^{\text{th}}$ qubit is a so-called ancilla qubit. Ancilla qubits can be viewed as “helper qubits” that are artificially introduced to make certain calculations feasible. Here, the ancilla qubit arises, since Σ is dilated to a unitary U_Σ in a higher-dimensional Hilbert space. In contrast to the two system qubits, it must always be initialized to $|0\rangle$. Besides U_Σ , two Hadamard gates are appended on the ancilla qubit, which prepare the system into a superposition of the basis states.

Applying the operators of the quantum circuit sequentially on the input state $|\psi\rangle \otimes |0\rangle$, the output state of the circuit can be calculated:

$$U H U_\Sigma V^\dagger H (|\psi\rangle \otimes |0\rangle) = \frac{1}{2} \begin{pmatrix} U(\Sigma_+ + \Sigma_-) V^\dagger |\psi\rangle \\ U(\Sigma_+ - \Sigma_-) V^\dagger |\psi\rangle \end{pmatrix} = \begin{pmatrix} M_k(t)|\psi\rangle \\ |\varphi\rangle \end{pmatrix}, \quad (4.10)$$

where $\Sigma_+ + \Sigma_- = 2\Sigma$, and thus $\frac{1}{2}U(\Sigma_+ + \Sigma_-)V^\dagger = M_k(t)$ as desired. The first half of the output vector is given by $M_k(t)|\psi\rangle$ and can be used to obtain $M_k(t)|\psi\rangle\langle\psi|M_k(t)$ by taking

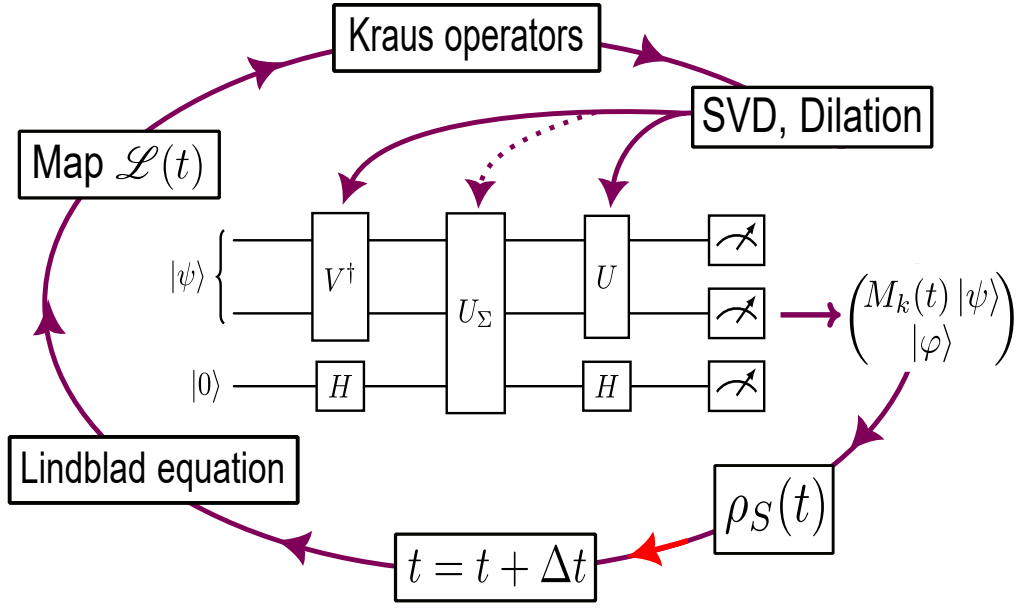


Figure 4.2: Illustration of the algorithm proposed by Seneviratne et al. to compute the dynamics of $\rho_S(t)$. For different time steps, the same procedure must be performed repeatedly, which is depicted by the purple ellipse. The algorithm starts at a specific time t and runs through all the steps until $\rho_S(t)$ is obtained. To advance in time, it updates t to the next step and repeats the entire cycle, as indicated by the red arrow.

the outer product with itself. The circuit must be repeated for each Kraus operator $M_k(t)$ at time t . Summing the results yields:

$$\rho_S(t) = \sum_k M_k(t) |\psi(0)\rangle \langle \psi(0)| M_k^\dagger(t) = \sum_k M_k(t) \rho_S(0) M_k^\dagger(t). \quad (4.11)$$

The second half of the output vector is given by $|\varphi\rangle$, and can be safely discarded. It is just a residual of the expansion of Σ to a higher dimension, and thus has no physical meaning. The circuit works for an arbitrary number of qubits, which must be adjusted to the dimensions of the system's Hilbert space. If the system Hilbert space is of dimension $2^n \times 2^n$, then $|\psi\rangle$ has a shape of $2^n \times 1$, $V^\dagger$ and U a shape of $2^n \times 2^n$, and U_Σ a shape of $2^{n+1} \times 2^{n+1}$.

Having completed all steps of the algorithm, a graphical summary is shown in Figure 4.2. In principle, the developed method is applicable to compute the time evolution of a multitude of open quantum systems, such as light-harvesting complexes, spin chains, or small molecular aggregates. In Seneviratne et al. [6], it is applied to the four-site Fenna-Matthews-Olsen complex.

4.1.4 Enhanced Algorithm

Ideally, the algorithm presented in the previous sections would be executed primarily on a quantum computer. However, a significant portion of the computational effort is performed classically, since the Kraus operators must be computed repeatedly for each time step. For each t , the quantum map $\mathcal{L}(t)$ is evaluated to construct the corresponding Choi matrix and derive the Kraus operators.

The vast majority of the calculation time for the classical algorithm could be spared if the Kraus operators only had to be calculated once for an arbitrary large time step Δt . The idea is to repeatedly apply the same circuit with a Kraus operator that propagates $\rho_S(t)$ to $\rho_S(t + \Delta t)$.

Thus, to propagate from 0 to 1000 femtoseconds with a time step of 10 femtoseconds, the same circuit must be applied $n_{\text{steps}} = 100$ times. Consequently, the computation executed by the classical routine must be performed only once, rather than n_{steps} times, which is quite a substantial improvement. Expressing this thought in a formula, the corresponding operator sum representation reads:

$$\rho_S(t + \Delta t) = \sum_k M_k(\Delta t) \rho_S(t) M_k^\dagger(\Delta t). \quad (4.12)$$

To realize the quantum circuit, we must find the correct state to initialize it for the next time step. Hence, we require a vector $|\psi'\rangle$ that corresponds to the density matrix $\rho_S(t)$, and thus fulfills $\rho_S(t) = |\psi'\rangle \langle\psi'|$. This condition implies that $\rho_S(t)$ must be pure. However, the interaction between the system and the environment leads to decoherence, which means that the initially pure state $\rho_S(0)$ becomes mixed. Thus, it is, in general, not possible to find a state vector $|\psi'\rangle$ that fulfills $\rho_S(t) = |\psi'\rangle \langle\psi'|$ and hence an alternative approach is required. To this end, we note that all information about the quantum state at time t is stored in the system density matrix, which can be correctly calculated for each time step. Therefore, we aim to use the system density matrix instead of the state vector to initialize the circuit. Since a quantum circuit can only be initialized with a vector, the system density matrix must be vectorized, where the length of $\text{vec}(\rho_S)$ depends on the dimension of the system. To implement $\text{vec}(\rho_S)$ instead of ρ_S , the operator sum representation must be vectorized:

$$\text{vec}(\rho_S(\Delta t)) = \text{vec}\left(\sum_k M_k(\Delta t) \rho_S(0) M_k^\dagger(\Delta t)\right) = \sum_k \left(M_k^*(\Delta t) \otimes M_k(\Delta t)\right) \text{vec}(\rho_S(0)). \quad (4.13)$$

Reviewing the circuit in Figure 4.1, we see that it is initialized with $|\psi\rangle$, and produces the relevant output $M_k(t) |\psi\rangle$, where the gates are obtained by a singular value decomposition of $M_k(t)$. Hence, by initializing the circuit with $|\text{vec}(\rho_S)\rangle$, we aim to obtain the relevant output state $(M_k^*(\Delta t) \otimes M_k(\Delta t)) |\text{vec}(\rho_S)\rangle$. Accordingly, it appears promising to perform

a singular value decomposition of $M_k^*(\Delta t) \otimes M_k(\Delta t)$ for each M_k , and sum the results to resemble Equation 4.13. We thus have:

$$M_k^*(\Delta t) \otimes M_k(\Delta t) = U \Sigma V^\dagger, \quad (4.14)$$

where U and $V^\dagger$ are already unitary, and Σ is non-unitary and diagonal. The corresponding circuit is shown in Figure 4.3, and a graphical summary of the algorithm in Figure 4.4.

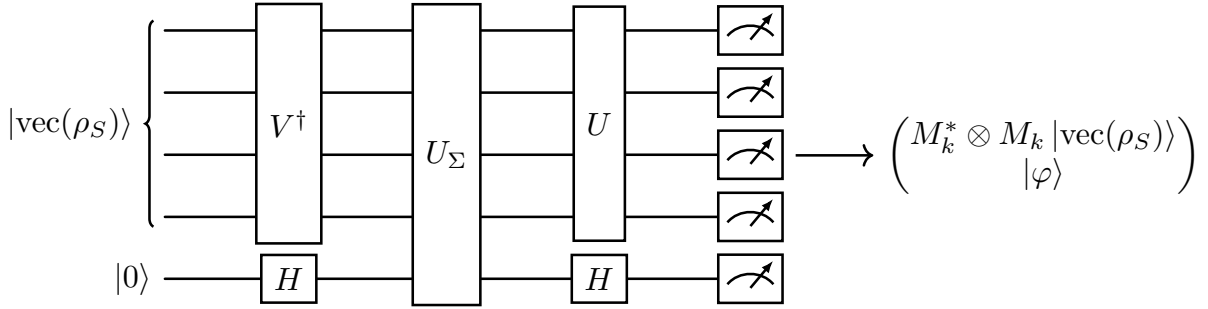


Figure 4.3: Quantum circuit for the implementation of the enhanced algorithm. To evolve the vectorized density matrix in time, a singular value decomposition of $M_k^*(\Delta t) \otimes M_k(\Delta t)$ is performed. Hence, compared to the algorithm presented by Seneviratne et al., more qubits are required. Using this circuit, the Kraus operators must be computed only once with the classical method, whereas the entire time evolution can be calculated on the quantum device.

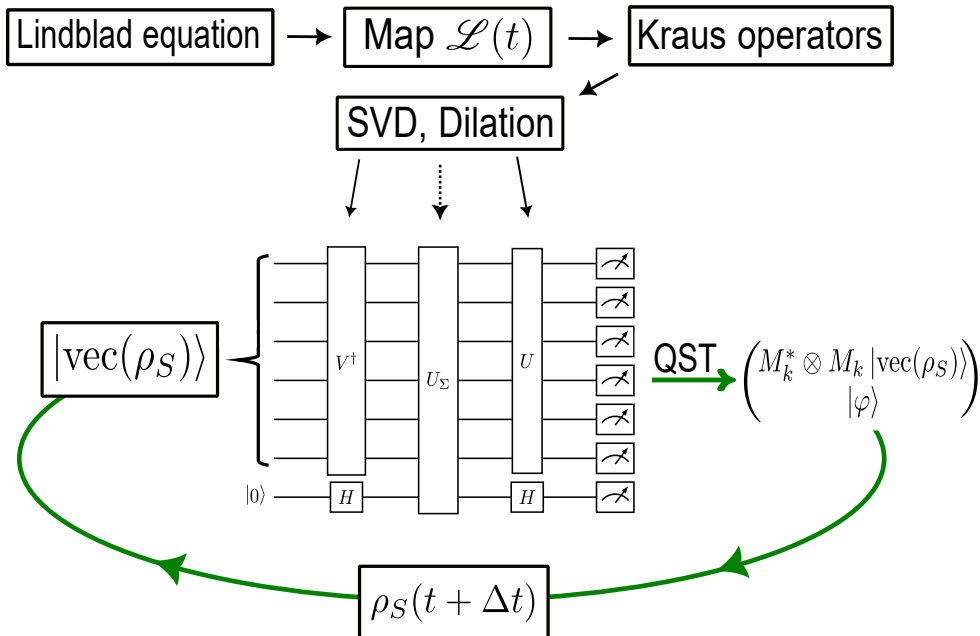


Figure 4.4: Graphical illustration of the enhanced algorithm. Instead of repeatedly calculating the Kraus operators, the circuit is only constructed once. To obtain $\rho_S(t + \Delta t)$, the vectorized ρ_S , computed in the previous time step, is reconstructed via QST and used as the initial state for the same circuit at the next time step, as indicated by the green ellipse.

4.2 Non-Markovian Dynamics

So far, the focus was limited to Markovian dynamics, and now turns to the non-Markovian case. A first approach, presented in Dan et al. [8], employs the hierarchical equations of motion (HEOM) to construct a non-Markovian object $G(t)$ that expresses the time evolution of open quantum systems in an operator sum representation. Similar to the method proposed by Seneviratne et al. [6], $G(t)$ is decomposed via a singular value decomposition and then mapped onto a quantum circuit, where it plays the role of a Kraus operator for the non-Markovian dynamics. In the HEOM-based formulation, however, the entire problem must first be solved classically. As a result, no computational advantage is obtained from involving a quantum device.

The method only demonstrates that non-Markovian dynamics, in which the bath retains memory of the system, can in principle be implemented on a quantum computer using HEOM. However, it does not permit the application of the enhanced algorithm and thus, developing a framework that combines HEOM with the enhanced algorithm for non-Markovian dynamics remains an open challenge.

An alternative non-Markovian method proposed by Seneviratne et al. [6] is introduced in Section 4.2.1. It is based on a path-integral formalism and leads to a quantum map $\mathcal{L}(t)$ from which the Kraus operators, capturing the non-Markovian effects, can be derived.

4.2.1 Enhanced Algorithm

In Seneviratne et al. [6], a tensor-network implementation of the path-integral approach was developed. Their method provides direct access to the non-Markovian quantum map $\mathcal{L}(t)$, which in turn allows the system density matrix to be propagated according to Equation 4.4. Consequently, the time evolution of the system density matrix can be calculated as follows:

$$\rho_s(s_t^+, s_t^-) = \int \mathcal{D}s^+ \int \mathcal{D}s^- \langle s_0^+ | \rho_0(0) | s_0^- \rangle \exp \left\{ \frac{i}{\hbar} (S[s^+] - S[s^-]) \right\} \text{IF}[s^+, s^-], \quad (4.15)$$

where

$$\text{IF}[s^+, s^-] = \exp \left\{ -\frac{1}{\hbar} \int_0^t dt' \int_0^{t'} dt'' [s^+(t') - s^-(t')] \right. \\ \left. [\alpha(t' - t'')s^+(t'') - \alpha^*(t' - t'')s^-(t'')] \right\} \quad (4.16)$$

is the Feynman-Vernon influence functional. The variables $s^+(t)$ and $s^-(t)$ label the ket- and bra-branches of the time contour, respectively, with the former evolving under U and the latter under $U^\dagger$. The expressions $S[s^+]$ and $S[s^-]$ are the action integrals for the free system

propagation, and $\langle s_0^+ | \rho_0(0) | s_0^- \rangle$ is the initial state. The nonlocal memory kernel $\alpha(t' - t'')$ captures the non-Markovianity [24].

Equation 4.15 defines an exact linear map $\mathcal{L}(t)$ that governs the non-Markovian dynamics of the system. Several numerically efficient methods have been proposed to implement this map, including quasi-adiabatic propagator path integrals, HEOM approaches developed by Kramer et al. [15][16], and the tensor network path-integral formalism (TNPI), which was used in Seneviratne et al. [6] to construct the linear map $\mathcal{L}(t)$, as can be seen in Figure 4.5.

Having obtained $\mathcal{L}(t)$ via TNPI, the procedure follows the steps introduced in Section 4.1. First, the corresponding Kraus operators, which capture the non-Markovian dynamics, are derived via the Choi decomposition. Then, the quantum circuit is constructed through a singular value decomposition and the required dilation. The enhanced algorithm can be applied using Equation 4.13 together with the circuit shown in Figure 4.3, where the Kraus operators M_k encode the non-Markovian behavior. Once the Kraus operators are obtained, the procedure should in principle be identical to the Markovian case in Figure 4.4, where the enhanced algorithm has already been validated.

We are therefore confident that the tensor network path integral approach extends the enhanced algorithm beyond Markovian dynamics, making it applicable to non-Markovian processes as well, although a formal proof has yet to be established.

Just as in the Markovian case, this approach should effectively transfer the majority of the computational workload from classical to the quantum routines.

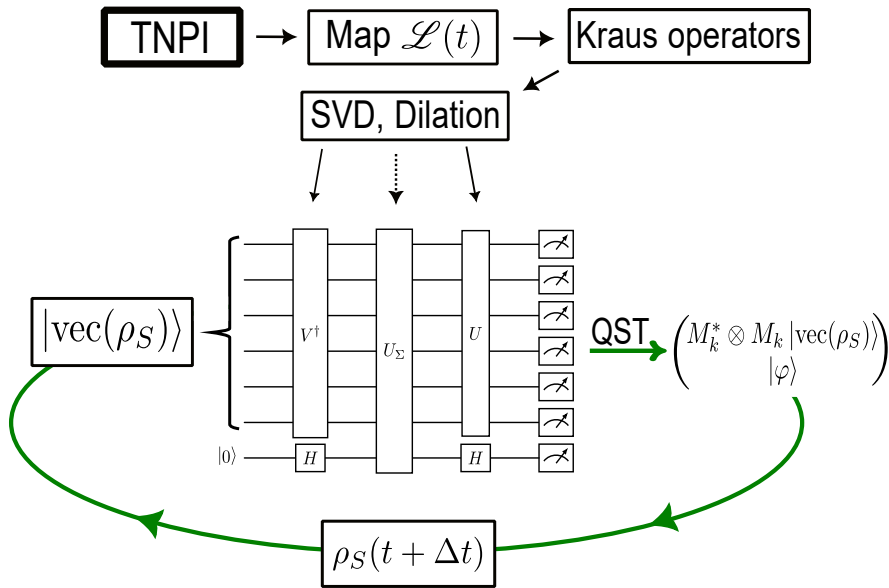


Figure 4.5: Illustration of the enhanced algorithm, where $\mathcal{L}(t)$ is obtained using TNPI instead of the vectorized Lindblad equation. The remaining procedure is equivalent to Figure 4.4.

4.3 Quantum State Tomography

This section discusses the challenges of quantum state tomography, including reconstructing states from density matrices, its resource-intensive scaling, and the demands imposed by the enhanced algorithm.

The output of the quantum circuit in Figure 4.3 is the vectorized system density matrix, which must be reshaped into its quadratic form. Unlike a state vector, tomography reconstructs a density matrix, yielding $|\text{vec}(\rho_S)\rangle \langle \text{vec}(\rho_S)|$ rather than $|\text{vec}(\rho_S)\rangle$. To recover the underlying state vector, one must perform an eigendecomposition and extract the eigenvector associated with the largest eigenvalue. If the reconstruction is accurate, this eigenvalue is close to 1, indicating that $|\text{vec}(\rho_S)\rangle \langle \text{vec}(\rho_S)|$ effectively represents a pure state.

However, in practice, additional errors are introduced. In the enhanced algorithm, the quantum state obtained at one time step is used as input for the following step. As a result, errors occurring at one stage propagate to the next, gradually accumulating and causing an increasing divergence. This highlights the need for quantum state tomography to be highly accurate in order to maintain a reliable time evolution. For this reason, I focus on evaluating different approaches and combining their strengths to achieve the highest possible accuracy.

Moreover, quantum state tomography is highly resource-intensive, requiring measurements across many bases. Since the no-cloning theorem forbids creating perfect copies of an unknown state, each measurement alters or destroys the system, meaning the entire time propagation must be recalculated to obtain another instance of the state. Thus, both repeated measurements and repeated full state generation are necessary, making the effort scale rapidly with system size.

In addition, the enhanced algorithm uses the vectorized density matrix as an input state, which enlarges the quantum circuit by two qubits and further increases computational costs. These limitations illustrate why quantum state tomography, although powerful, is still far from being a practical solution for larger systems. Overcoming these challenges in terms of error accumulation, scalability and measurement overhead remains an important task for future work.

5 Research Questions and Hypotheses

The advanced algorithm developed in my Master’s thesis fundamentally relies on quantum state tomography, since the full state must always be reconstructed for the method to work in practice. Establishing that this reconstruction process can be carried out efficiently and robustly is therefore the first step.

To this end, I will explore three complementary families of reconstruction techniques and combine their strengths to yield a scheme that is both scalable and resilient under realistic experimental conditions. Once the combined method for state reconstruction has been demonstrated to work reliably, the next step is to extend the approach to non-Markovian dynamics.

In my Master’s thesis, I showed that the approach proposed by Dan et al. [8] using HEOM is not suited for the enhanced algorithm and heavily relies on classical pre-processing. Hence, concerning HEOM, a new method must be developed. However, using the tensor network path integral formalism proposed in Seneviratne et al. [6], we are very confident that the enhanced algorithm can shift the majority of the computational workload towards the quantum computer, although a strict proof remains an open task.

5.1 Compressed Sensing Tomography

Many relevant quantum states, including typical outputs of quantum circuits, are pure or nearly pure, which implies that their density matrices contain only a few significant eigenvalues compared with the full Hilbert space dimension. This rank sparsity can be exploited to reduce the number of measurements, since the reconstruction effort can focus on the essential components of the state rather than attempting to recover the entire density matrix [20].

- **Idea:** Recover an accurate low-rank approximation of the full density matrix by targeting its dominant eigencomponents.
- **Question:** How robust are low-rank reconstructions under realistic noise and repeated propagation steps?
- **Realization:** Choose a structured set of measurements that is informative for reconstructing the dominant components.
- **Impact:** Accurate low-rank approximations make iterative state propagation feasible in practice.

For states with pronounced rank sparsity, low-rank reconstruction offers a scalable alternative to full tomography by substantially reducing the measurement burden and enabling efficient iterative propagation [11].

5.2 Matrix Product State Tomography

Matrix product state tomography relies on the observation that many relevant quantum states are only weakly entangled. This allows the global state to be represented by local tensors, each associated with a single site. A local tensor is a small multidimensional array that links a site to its neighbors and captures the local amplitudes and short-range correlations. By combining all tensors, one obtains an efficient reconstruction of the full wave function or density matrix [25].

- **Idea:** Recover the global state by fitting local tensors from a set of local reduced density matrices on adjacent blocks.
- **Question:** How robust is the reconstruction under experimental noise, residual long-range correlations, and bond growth during the time evolution?
- **Realization:** Measure reduced density matrices for adjacent blocks and reconstruct tensors using local tomography or variational tensor-network optimization.
- **Impact:** Due to its polynomial rather than exponential scaling with system size, the required measurement effort is reduced substantially.

In summary, matrix product state tomography offers a powerful alternative to full state reconstruction whenever entanglement remains limited. By exploiting the tensor-network structure, it turns an otherwise intractable task into a feasible method for characterizing open quantum systems of modest size [10].

5.3 Neural-Network-Based Tomography

A third direction will investigate machine-learning methods to represent quantum states using only a limited set of measurements. These approaches aim to compress the essential information of the quantum state into a compact model that generalizes to unmeasured observables, making them especially promising for large systems where conventional reconstruction becomes impractical. Generative and autoregressive architectures are particularly well-suited to capture the structure of the states produced by the circuits of interest.

- **Idea:** Represent the quantum state as a compact model that can predict relevant observables from limited measurement data.
- **Question:** How reliably can the model generalize to unmeasured observables with finite training data?
- **Realization:** Train generative models on selected measurements, enforce physical constraints, and quantify prediction uncertainty for safe use in iterative propagation.

- **Impact:** A powerful machine-learning model that enables efficient reconstruction of large quantum states for systems with up to 100 qubits.

By compressing high-dimensional quantum states into compact, predictive models, this machine-learning approach can dramatically reduce the measurement burden while still providing the information required for iterative state propagation [12][18].

5.4 Practical Realization of Non-Markovian Maps on Quantum Devices

All three quantum state reconstruction techniques will be developed with the same integration objective: the reconstructed state must be directly usable as the input for the next step of the quantum propagation. Extending these methods to non-Markovian dynamics, where the system evolution depends on its history, requires designing effective maps that capture memory effects and can be executed on quantum hardware. This raises several important questions:

- Which effective non-Markovian maps exist that can be realized on quantum hardware, and what is the associated computational cost for each approach?
- How can these maps be dilated into circuits with manageable depth and gate counts?

To address these challenges, I will focus on two complementary methods for simulating non-Markovian dynamics on quantum devices: tensor network path integral methods (TNPI) and the hierarchical equations of motion (HEOM).

HEOM provides a numerically exact description of non-Markovian open quantum system dynamics. In my Master's thesis, I demonstrated that the HEOM-based approach proposed by Dan et al. [8] is not directly compatible with the enhanced algorithm and relies heavily on classical pre-processing. Therefore, extending the enhanced algorithm using HEOM requires the development of a new method that can accurately integrate quantum state propagation while minimizing classical computational overhead. Here I will benefit from the expertise of Univ.-Prof. Dr. Tobias Kramer, who has several publications concerning the efficient implementation of HEOM [15][16].

On the other hand, tensor network path integral methods, as proposed by Seneviratne et al. [6], provide a framework for constructing the non-Markovian Kraus operators. The main goal is to rigorously demonstrate that the enhanced algorithm can be successfully implemented within this framework. By leveraging this approach, the vast majority of the computational workload can be shifted onto the quantum computer, offering an efficient path to simulate non-Markovian dynamics once the method's feasibility has been formally demonstrated.

In short, the two central goals concerning non-unitary dynamics are to develop a new algorithm using HEOM and to validate the enhanced algorithm for non-Markovian dynamics with the tensor network path integral (TNPI) approach.

Given the limitations of current quantum processors, such as finite coherence times and non-negligible gate errors, particular care will be taken to compress circuits and minimize gate depth. These measures will help ensure that the full workflow can be executed efficiently on near-term quantum hardware.

5.5 Validation Strategy and Anticipated Outcomes

The proposed program can be validated on a range of open quantum systems, such as the Fenna–Matthews–Olson light-harvesting complex, spin chains, or small molecular aggregates, where the dynamics are well understood and can serve as benchmarks. Validation will focus on assessing both the accuracy and the practical feasibility of the reconstruction and propagation methods:

- Assess scaling, robustness and accuracy through numerical simulations with and without noise.
- Demonstrate feasibility with experiments on available quantum hardware.
- Quantify performance using metrics such as number of measurements, computational time and memory, circuit depth, gate counts, and fidelity over multiple propagation steps.

Through this validation, the project will develop practical reconstruction strategies that reduce measurement and computational overhead, clarify which parts of the workflow can be executed directly on quantum processors, and provide a clear assessment of the remaining challenges for simulating non-Markovian dynamics on near-term quantum devices.

6 Work Packages and Schedule

In the following, I outline the planned work packages and the proposed schedule for the three-year project. The overall allocation of topics is illustrated in the Gantt chart in Figure 6.1. What follows is a compact overview of the four research packages that form the core of the proposed work. Each package is presented with a brief subtitle and a straightforward explanation of its goals, the methods to be used, and the main challenges to be addressed.

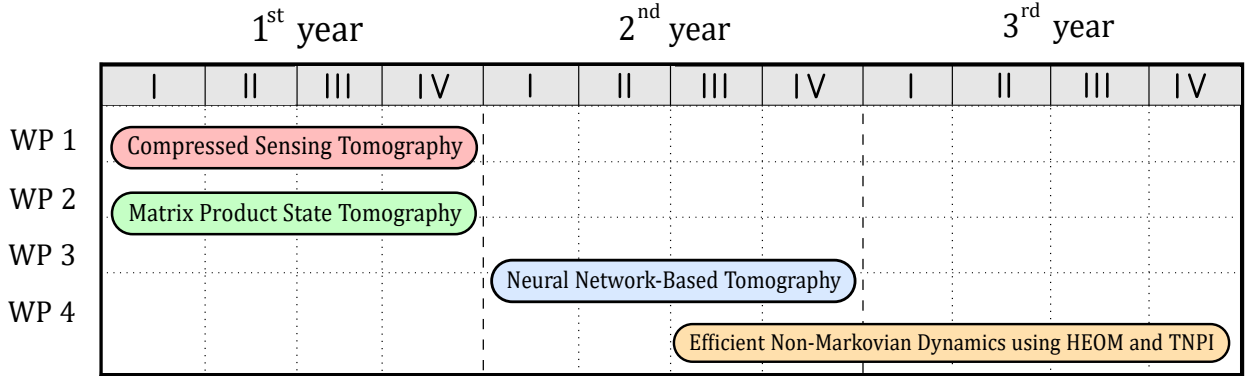


Figure 6.1: Gantt chart showing the planned schedule for the four work packages over the proposed three-year project period. Each bar indicates the expected timeframe for completing the respective research package.

WP 1: Compressed Sensing Tomography for Low-Rank Matrices

I plan to begin my research with compressed sensing tomography, as it is particularly well-suited for low-rank matrices. Since quantum circuits naturally output pure states, which are rank-1, the approach is expected to readily deliver good results for systems of moderate size.

WP 2: Matrix Product State Tomography

In parallel to WP 1, I aim to investigate matrix product state tomography, which is particularly effective when states exhibit only limited entanglement. Since many open quantum systems produce such states, this method is expected to provide an efficient reconstruction of quantum systems beyond the reach of full tomography.

WP 3: Neural-Network Quantum State Tomography

WP 3 focuses on reconstructing large quantum states with up to 100 qubits using machine learning techniques. Building on the gained knowledge during my AI Master's, my goal is to learn compact representations of quantum states from limited samples by training autoregressive and variational neural-network models on measurement data.

WP 4: Efficient Simulation of Non-Markovian Dynamics

In the last phase, I aim to develop an improved non-Markovian algorithm using HEOM and to validate the enhanced algorithm for the tensor network path integral approach. In addition, to enable the planned simulations on state-of-the-art quantum devices, particular attention will be paid to find efficient circuit representations to minimize the gate count.

The four work packages are designed to complement each other, as is demonstrated in Reference [26] (2024) for neural-network-based tomography and compressed sensing tomography. In general, WP 1 implements low-rank reconstruction methods for nearly pure quantum states, WP 2 develops matrix product state tomography to efficiently reconstruct weakly entangled states, and WP 3 provides flexible, data-driven models to scale up to larger systems. All three approaches are particularly well-suited for open quantum systems, where the states often exhibit limited entanglement or rank structure. Together, the work packages aim to reduce measurement and computational effort while producing reliable state reconstructions that can be used in later stages, including non-Markovian dynamics and tests on real quantum hardware planned for the second half of the project.

7 Requested Funding Period

The proposed project is ambitious and covers several interrelated work packages, from low-rank reconstruction methods and matrix product state Tomography to neural-network-based approaches and non-Markovian dynamics. Each of these steps requires careful development, testing, and validation.

Since I will start the project directly after completing my Master's, the first phase will focus on building the necessary expertise in quantum tomography techniques and machine learning methods. Many of the methods are iterative and cumulative, meaning that early results feed into later work packages. The project also involves testing on real quantum hardware, which takes additional time due to hardware constraints and optimization needs.

Considering the complexity of the project, I propose a three-year funding period to move beyond proof-of-concept algorithms toward applications on real quantum hardware. The stepwise completion of the four work packages will lay the foundation to develop a promising algorithm for full state reconstruction, paving the way to propagate the dynamics of open quantum systems almost entirely on quantum devices.

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